BRAC University (Department of Computer Science and Engineering)
CSE 221 (Algorithms) for Summer 2026 Semester
Quiz 1 - Set A
Full Marks: 15 | Time: 30 minutes

Name:_____**ID:**_____**Section:**_____

1. Write down the time complexity of the following code snippet in Big-O notation. Here, n is an input variable and k is a constant (k>1). **Show your work.** [2.5 marks]

```
for (int i = 1; i <= n; i += k) {
    int sum = 0;

    for (int j = n; j >= 1; j = j / k) {
        for (int m = 1; m <= n; m += 2) {
            if (m % 2 == 0) {
                break;
            }

            sum = sum + 10;
        }
    }
}
```

2. Explain why **Big-O notation** alone may not be enough to fully describe the efficiency of an algorithm. [2.5 marks]

3. Given the following array:

4	9	15	22	31	40

Simulate **ternary search** to find the target value 18. Show all steps clearly. [5 marks]

4. Consider a sorting algorithm that repeatedly compares adjacent elements and swaps them if they are in the wrong order. After each full pass, the largest unsorted element moves to its correct position.
Identify the algorithm. Then suggest an improvement that allows the algorithm to stop early if the array becomes sorted before all passes are completed. **Write the pseudocode for the improved version and analyze its best-case and worst-case time complexities.** [5 marks]

Start writing from here

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Quiz 2
Full Marks: 20 | Time: 35 minutes

Name:_____ **ID:**_____ **Section:**_____

1. Given the following recurrence relation: **4**

$$T(n) = T(3n/4) + T(n/3) + \Theta(n)$$

The two recursive branches reduce the problem size at different rates. Determine which branch produces the longest path in the recursion tree and derive the height of that branch.

2. You are given three sorted arrays A, B, and C of lengths n1, n2, and n3, respectively. Each array is sorted in non-increasing order. Write the pseudocode/code for a function 3-Merge (A, B, C) that merges these three arrays into a single sorted array D of length n = n1 + n2 + n3. Your algorithm must run in O(n) time. **6**

3. Given the following array (A): **10**

-5	4	-1	3	-2	6	-7	2

a. Simulate the divide-and-conquer maximum-subarray algorithm on array (A). During the simulation, clearly highlight every instance where the Crossing Sum (CSS) is greater than both the Left Sum (LSS) and the Right Sum (RSS).
b. Then, write the recurrence relation for the divide-and-conquer approach and use the Master Theorem to determine its time complexity. **Show all necessary steps.**

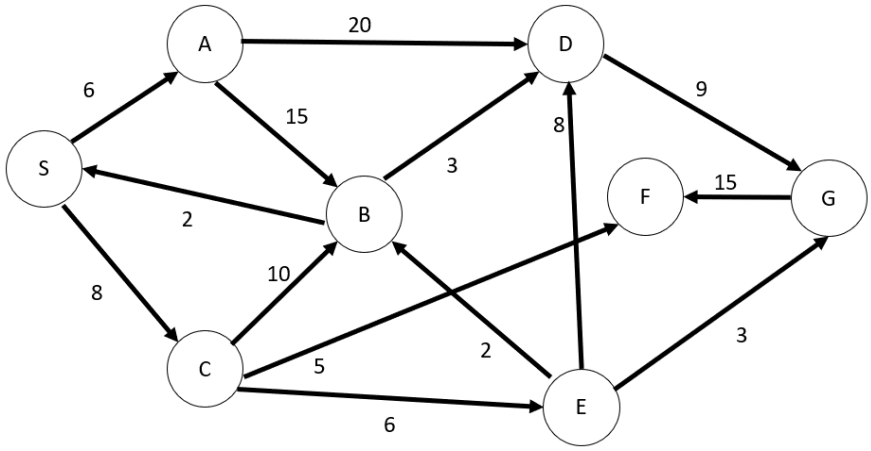
Start writing from here

1. Write down one usecase/application of the end times in DFS. **2**

2. Compare the time complexity of Dijkstra's algorithm when implemented using an array and a priority queue. Briefly explain why their running times differ. **5**

3. A graph always has a unique topological sorting. Do you agree or disagree? Explain your reasoning. **3**

4. **10**



Using S as the source vertex, simulate Dijkstra's algorithm to determine the shortest distance to every other vertex. **List all edges where relaxation updates a distance.**

Start writing from here

Welcome to the Cyber Carnival, where you can earn prize money by completing different virtual challenges. You have only 10 hours available. Each challenge requires a specific amount of time and awards a fixed amount of prize money when completed. A challenge must be completed fully to earn its prize, and each challenge can be attempted at most once. Partial completion provides no prize.

Challenge	Required Time	Prize Money
Neon Racer	7 hours	$75
Metro Outlaw	10 hours	$90
Cyber Agent	2 hours	$30
Zombie Mansion	3 hours	$55
Jungle Assault	1 hour	$17
Shadow Warrior	5 hours	$47

Identify a suitable algorithmic approach to maximize the total prize money earned within 10 hours. Show a complete simulation of your algorithm and determine the selected challenges, and the maximum prize money earned. **[20 Marks]**

Start writing from here

1. A software project contains seven tasks: 10

   V = {A, B, C, D, E, F, G}

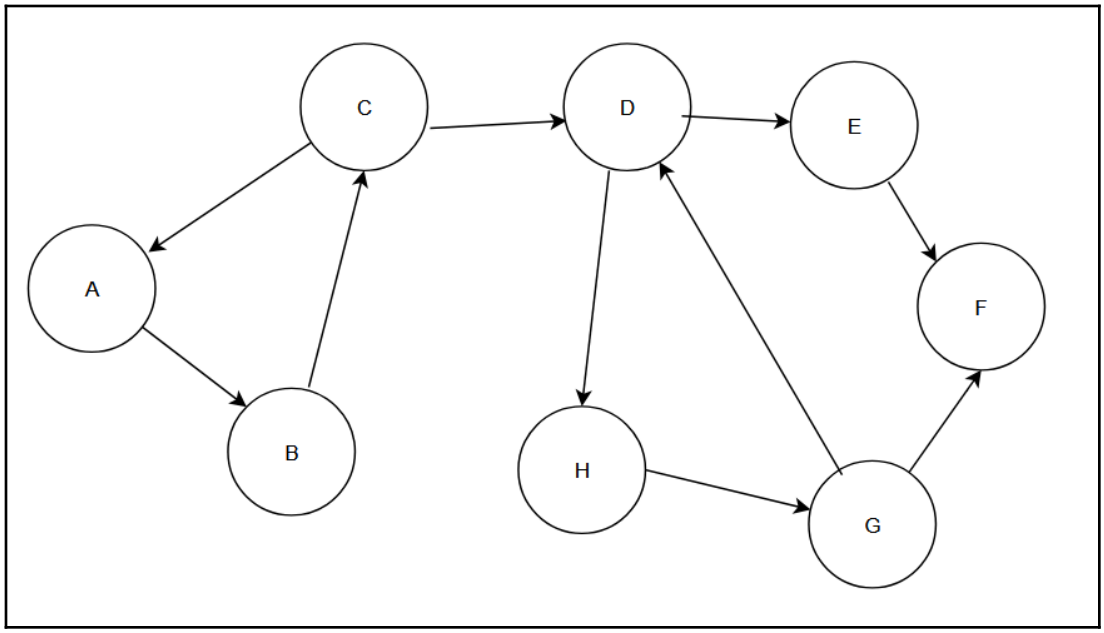
   The dependencies between the tasks are represented by the following directed edges:

   A → C, A → D, B → D, B → E, C → F, D → F, E → G, F → G

   An edge X → Y means that task X must be completed before task Y.

Using a suitable algorithm, determine a valid order in which all tasks can be completed. Show the complete simulation of the algorithm, and process tasks alphabetically whenever multiple tasks are available.

2. Find all the strongly connected components of the given directed graph. 10



Show the complete simulation of the selected algorithm